

— CS1006 – Data Structures —

§ 200+ MCQs with Answers

• *Unit I: Abstract Data Types & Memory Allocation | Unit II: Linked List*

Side Remark

An array behaves like numbered pages. A linked list behaves like following references from page to page.

§ SECTION A — Unit I: Data Structures Basics & ADT

1. Which of the following best defines a data structure?

- a) A programming language
- b) A particular way of organizing data in a computer so that it can be used efficiently ✓
- c) A type of algorithm
- d) A memory management technique

2. The formula "Data Structures + Algorithms = ____" is attributed to which concept?

- a) Software Engineering
- b) Program ✓
- c) Database
- d) Operating System

3. Which of the following is NOT a primitive data type?

- a) int
- b) char
- c) float
- d) Linked List ✓

4. A Non-Primitive Data Structure can be constructed with the help of:

- a) Only arrays
- b) Only pointers
- c) Any primitive data structure ✓
- d) Only integers

5. Which of the following is a Linear Data Structure?

- a) Tree
- b) Graph
- c) Queue ✓

Reader's Note

Stack is quick and scoped. Heap is flexible and manual. In C, memory mistakes are rarely quiet.

- d) Heap

6. Which of the following is a Non-Linear Data Structure?

- a) Stack
- b) Queue
- c) Array
- d) Tree ✓

7. In a Linear Data Structure, how many data elements can be directly reached at a time?

- a) All elements
- b) Two elements
- c) One element ✓
- d) Three elements

8. Which operation adds a new data element to a data structure?

- a) Traversal
- b) Deletion
- c) Insertion ✓
- d) Merging

9. Arranging data elements in a specified order is called:

- a) Merging
- b) Sorting ✓
- c) Searching
- d) Traversal

10. Combining elements of two similar data structures to form a new structure of the same type is:

- a) Sorting
- b) Merging ✓
- c) Searching
- d) Insertion

11. An Abstract Data Type (ADT) is:

- a) A concrete implementation of data
- b) A mathematical model of a data structure ✓
- c) A type of array
- d) A hardware component

12. ADT encapsulates a range of data values and their:

- a) Addresses

- b) Pointers
- c) Functions ✓
- d) Algorithms

13. Which of the following is an example of an ADT?

- a) Integer
- b) Float
- c) Stack ✓
- d) Binary digit

14. The List ADT represents:

- a) A non-linear collection of elements
- b) A countable number of ordered values ✓
- c) An unordered set of unique values
- d) A circular buffer

15. If $N = 0$, the list is considered:

- a) Full
- b) Empty ✓
- c) Circular
- d) Sorted

16. Which List ADT operation returns the number of elements in the list?

- a) `get()`
- b) `isEmpty()`
- c) `size()` ✓
- d) `isFull()`

17. Which List ADT operation removes the first occurrence of an element from a non-empty list?

- a) `removeAt()`
- b) `delete()`
- c) `remove()` ✓
- d) `pop()`

18. Which List ADT operation replaces an element at any position?

- a) `update()`
- b) `replace()` ✓
- c) `swap()`
- d) `modify()`

19. Two standard implementations for the List ADT are:

- a) Stack and Queue
- b) Array-based and Linked list ✓
- c) Tree and Graph

- d) Hash table and Heap
- 20.** An array is a _____ access data structure.
- a) Sequential
 - b) Random ✓
 - c) Linear
 - d) Circular
-

§ SECTION B — Memory Layout in C Programs

21. When a C program is compiled, which type of file is created?

- a) .txt file
- b) .pdf file
- c) Binary executable file (.exe) ✓
- d) .obj file only

22. How many components does the memory layout of a C program have?

- a) Four
- b) Five
- c) Six ✓
- d) Seven

23. Which memory segment stores the compiled binary instructions of a program?

- a) Stack
- b) Heap
- c) Data Segment
- d) Text Segment ✓

24. The Text Segment has which type of permission?

- a) Write-only
- b) Read-write
- c) Read-only ✓
- d) Execute-only

25. Which segment stores the values of global, static, external, and initialized constant variables?

- a) BSS Segment
- b) Stack
- c) Heap
- d) Initialized Data Segment ✓

26. The Initialized Data Segment has:

- a) Read-only permission
- b) Write-only permission

- c) Read-write permission ✓
- d) No permission

27. A const variable in C is stored in which area of the Data Segment?

- a) Read-write area
- b) Stack
- c) Read-only area ✓
- d) Heap

28. The Uninitialized Data Segment is also known as:

- a) TSS
- b) BSS (Block Started by Symbol) ✓
- c) CSS
- d) DSS

29. Every data in BSS is initialized to arithmetic _____ before program execution.

- a) 1
- b) -1
- c) 0 ✓
- d) NULL (for non-pointers)

30. Stack memory behaves as a:

- a) FIFO buffer
- b) LIFO buffer ✓
- c) Random access buffer
- d) Circular buffer

31. Stack memory stores the value of:

- a) Global variables
- b) Local variables and function parameters ✓
- c) Dynamically allocated memory
- d) Constants only

32. Stack grows in which direction relative to heap?

- a) Same direction
- b) Opposite direction ✓
- c) Perpendicular
- d) It doesn't grow

33. Heap memory is allocated during:

- a) Compile time
- b) Link time
- c) Run time ✓
- d) Load time

34. Heap is also referred to as:

- a) Static memory
- b) Free pool of memory ✓
- c) Text segment
- d) BSS

35. How many functions are used to manage heap memory in C?

- a) Two
- b) Three
- c) Four ✓
- d) Five

36. Which error occurs when a program tries to access a memory segment incorrectly?

- a) Stack overflow
- b) Null pointer exception
- c) Segmentation fault ✓
- d) Bus error

37. The Text Segment in RAM is:

- a) Private to each process
- b) Shareable ✓
- c) Write-protected and not shareable
- d) Only readable once

38. BSS also contains:

- a) Initialized local variables
- b) Static and global variables initialized to 0 ✓
- c) Constants
- d) Code instructions

§ SECTION C — Memory Allocation (Static & Dynamic)

39. Compile-time memory allocation is also referred to as:

- a) Dynamic memory allocation
- b) Static or stack memory allocation ✓
- c) Heap memory allocation
- d) Virtual memory allocation

40. In static memory allocation, once memory is allocated:

- a) It can be resized with realloc()
- b) Its size cannot be changed ✓
- c) It is freed automatically
- d) It moves to heap

41. Which of the following is a Key Feature of Static Memory Allocation?

- a) Allocation done at run time
- b) Memory size can be changed
- c) Execution is faster than dynamic memory allocation ✓
- d) Uses heap data structure

42. Static memory allocation uses which data structure?

- a) Queue
- b) Heap
- c) Stack ✓
- d) Tree

43. In static memory allocation, memory is allocated:

- a) After runtime
- b) Before runtime ✓
- c) During garbage collection
- d) On demand

44. Dynamic memory allocation is also known as:

- a) Stack allocation
- b) BSS allocation
- c) Heap allocation / free pool allocation ✓
- d) Text allocation

45. Which is the only way to access memory on the heap?

- a) Arrays
- b) Pointers ✓
- c) Functions
- d) Structures

46. The malloc() function stands for:

- a) Memory algorithm
- b) Memory allocation ✓
- c) Multiple allocation
- d) Main allocation

47. What does malloc() return?

- a) An integer
- b) A pointer of type void ✓
- c) A pointer of type int
- d) A null pointer always

48. What initial value does malloc() assign to the allocated memory?

- a) Zero
- b) One
- c) Garbage value ✓

- d) NULL

49. Given: `ptr = (int*) malloc(100 * sizeof(int));` and `sizeof(int) = 4`, how many bytes are allocated?

- a) 100
- b) 4
- c) 400 ✓
- d) 104

50. The correct syntax of `malloc()` is:

- a) `ptr = malloc(n, size)`
- b) `ptr = (cast-type*) malloc(byte-size)` ✓
- c) `ptr = calloc(byte-size)`
- d) `ptr = realloc(byte-size)`

51. `calloc()` stands for:

- a) Calculated allocation
- b) Contiguous allocation ✓
- c) Circular allocation
- d) Complete allocation

52. What value does `calloc()` use to initialize each block of allocated memory?

- a) Garbage value
- b) 1
- c) -1
- d) 0 ✓

53. How many parameters does `calloc()` take?

- a) One
- b) Two ✓
- c) Three
- d) Four

54. The correct syntax of `calloc()` is:

- a) `ptr = (cast-type*) calloc(byte-size)`
- b) `ptr = (cast-type*) calloc(n, element-size)` ✓
- c) `ptr = calloc(ptr, newSize)`
- d) `ptr = (cast-type*) malloc(n, element-size)`

55. `ptr = (float*) calloc(25, sizeof(float));` allocates space for how many float elements?

- a) 4
- b) 100
- c) 25 ✓
- d) 50

56. `realloc()` is used to:

- a) Allocate new memory
- b) Free all memory
- c) Dynamically change the memory allocation of a previously allocated memory ✓
- d) Initialize memory to zero

57. When `realloc()` expands memory and adjacent space is not available, it:

- a) Returns NULL
- b) Allocates at a new location ✓
- c) Throws an exception
- d) Reduces memory instead

58. New blocks allocated by `realloc()` are initialized with:

- a) Zero
- b) One
- c) Garbage value ✓
- d) Previous values

59. The correct syntax of `realloc()` is:

- a) `realloc(n, newSize)`
- b) `realloc(ptr, n)`
- c) `ptr = realloc(ptr, newSize)` ✓
- d) `ptr = realloc(newSize)`

60. The `free()` function is used to:

- a) Allocate memory on heap
- b) Resize memory
- c) De-allocate dynamically allocated memory ✓
- d) Initialize memory to zero

61. Memory allocated using `malloc()` or `calloc()` is NOT de-allocated on its own. Which function handles this?

- a) `malloc()`
- b) `realloc()`
- c) `calloc()`
- d) `free()` ✓

62. The syntax of `free()` is:

- a) `free(ptr, size)`
- b) `free(ptr)` ✓
- c) `free(size)`
- d) `free()`

63. Which of the following is a feature of Dynamic Memory Allocation?

- a) No reusability
- b) Execution is faster

- c) Memory is allocated before runtime
- d) Can reuse allocated memory after releasing with free() ✓

64. Which function requires library functions like malloc(), calloc(), or realloc()?

- a) Compile time memory allocation
- b) Static memory allocation
- c) Run time memory allocation ✓
- d) Stack allocation

65. In compile-time memory allocation, can the memory size be changed during runtime?

- a) Yes, using realloc()
- b) Yes, using malloc()
- c) No ✓
- d) Yes, using free()

66. Which allocation type requires you to know exactly how many bytes are needed in advance?

- a) Run time allocation
- b) Compile time allocation ✓
- c) Heap allocation
- d) Virtual allocation

67. Which type of memory allocation has no memory wastage?

- a) Static allocation
- b) Compile time allocation
- c) Dynamic allocation ✓
- d) Text segment allocation

68. The sizeof operator in C is used to:

- a) Allocate memory
- b) Free memory
- c) Determine memory occupied by a data type ✓
- d) Assign values to variables

§ SECTION D — Arrays

69. An array can store a fixed-size sequential collection of elements of:

- a) Different types
- b) The same type ✓
- c) Pointers only
- d) Characters only

70. How are elements in an array accessed?

- a) By value

- b) By name
- c) By index ✓
- d) By pointer only

71. All arrays consist of _____ memory locations.

- a) Scattered
- b) Non-contiguous
- c) Contiguous ✓
- d) Circular

72. The lowest address in an array corresponds to:

- a) The last element
- b) The middle element
- c) The first element ✓
- d) A random element

73. To check if a list is full, which condition is verified?

- a) Current Size == 0
- b) Current Size == Max Size ✓
- c) Current Size > Max Size
- d) Current Size < 0

74. To check if a list is empty, which condition is verified?

- a) Current Size == Max Size
- b) Current Size < 0
- c) Current Size == 0 ✓
- d) Current Size == 1

75. When inserting an element at the end of a list, the new position is:

- a) Current Size
- b) Current Size - 1
- c) Current Size + 1 ✓
- d) 0

76. After inserting an element at the end, the Current Size is:

- a) Decreased by 1
- b) Remains unchanged
- c) Increased by 1 ✓
- d) Reset to 0

77. When deleting from the end of the list, the position of the element to delete is:

- a) 0
- b) Current Size - 1
- c) Current Size ✓
- d) Max Size

78. When inserting at the front of a list, all existing elements must be shifted:

- a) One position backward
- b) One position forward ✓
- c) Two positions forward
- d) No shifting needed

79. When deleting from the front of a list, remaining elements are moved:

- a) One position forward
- b) One position backward ✓
- c) Two positions backward
- d) No shifting needed

80. If the nth position to insert is greater than the current size + 1, what happens?

- a) The element is inserted at position 1
- b) The element is inserted at the end
- c) An error message is returned ✓
- d) The list is cleared

81. For the Search operation in a list, what is returned if an element is found?

- a) The value of the element
- b) True
- c) The position of the element ✓
- d) A pointer

82. The Traversal operation processes all data elements in a data structure:

- a) Randomly
- b) Exactly once ✓
- c) Multiple times
- d) In reverse order only

§ SECTION E — Unit II: Linked Lists

83. A Linked List is a _____ access data structure.

- a) Random
- b) Sequential ✓
- c) Direct
- d) Circular

84. Unlike arrays, linked list elements are stored at:

- a) Contiguous memory locations
- b) Non-contiguous locations, linked via pointers ✓
- c) Fixed memory addresses
- d) The stack only

85. Which of the following is a limitation of arrays that motivates use of linked lists?

- a) Arrays are too slow to traverse
- b) The size of arrays is fixed ✓
- c) Arrays cannot store integers
- d) Arrays do not support indexing

86. Which of the following is an advantage of a linked list over an array?

- a) Allows random access
- b) Uses less memory
- c) Dynamic size ✓
- d) Faster binary search

87. Which of the following is a drawback of a linked list?

- a) Dynamic size
- b) Ease of insertion/deletion
- c) Random access is not allowed ✓
- d) Supports sequential traversal

88. Each node in a singly linked list consists of:

- a) Data only
- b) Pointer only
- c) Data and pointer to the next node ✓
- d) Data and two pointers

89. In C, a linked list node is typically represented using:

- a) Arrays
- b) Unions
- c) Structures ✓
- d) Enumerations

90. The first node in a linked list is called:

- a) Tail
- b) Head ✓
- c) Root
- d) Parent

91. If a linked list is empty, the value of head is:

- a) 0
- b) 1
- c) NULL ✓
- d) -1

92. Which type of linked list has each node pointing only to the next node?

- a) Doubly Linked List
- b) Circular Linked List

- c) Singly Linked List ✓
- d) Circular Doubly Linked List

93. A Doubly Linked List node contains:

- a) Data and one pointer
- b) Data and two pointers (prev and next) ✓
- c) Data and three pointers
- d) Only pointers

94. In a Circular Singly Linked List, the last node points to:

- a) NULL
- b) The previous node
- c) The head node ✓
- d) Itself

95. A Circular Doubly Linked List combines features of:

- a) Stack and Queue
- b) Doubly Linked List and Circular Linked List ✓
- c) Array and Stack
- d) Singly and Doubly Linked List only

96. Which linked list allows traversal in both forward and backward directions?

- a) Singly Linked List
- b) Circular Singly Linked List
- c) Doubly Linked List ✓
- d) None

97. Binary search cannot be performed efficiently on linked lists because:

- a) Linked lists cannot store integers
- b) Random access is not allowed ✓
- c) Linked lists are too large
- d) Pointers slow down access

98. In a Singly Linked List, which direction of traversal is possible?

- a) Backward only
- b) Both forward and backward
- c) Forward only ✓
- d) Neither

99. What is the extra memory requirement specific to linked lists?

- a) Extra data field for each node
- b) Extra pointer for each element ✓
- c) Extra array index for each node
- d) Extra stack frame

100. Lists form the basis for which other abstract data types?

- a) Trees and Graphs
 - b) Queue and Stack ✓
 - c) Heaps and Hash Tables
 - d) Binary Trees only
-

§ SECTION F — Singly Linked List Operations

101. To insert a node at the beginning of a singly linked list, the new node's next should point to:

- a) NULL
- b) The current tail
- c) The current head ✓
- d) The second node

102. After inserting at the beginning, the new node becomes:

- a) The tail
- b) A middle node
- c) The new head ✓
- d) Unchanged

103. To insert at the end of a singly linked list, we traverse until:

- a) head == NULL
- b) temp == NULL
- c) temp->next == NULL ✓
- d) temp->data == 0

104. To delete the front node of a singly linked list, the new head should be set to:

- a) NULL
- b) The second node (head->next) ✓
- c) The last node
- d) head->prev

105. After deleting the front node, which function is called on the old head?

- a) malloc()
- b) calloc()
- c) realloc()
- d) free() ✓

106. To delete the last node of a singly linked list, we need to:

- a) Set head = NULL
- b) Traverse to the second-to-last node and set its next to NULL ✓
- c) Traverse to the last node and set its data to 0
- d) Set head->next = head

107. In the search operation on a singly linked list, what does the variable found indicate?

- a) Whether the list is empty
- b) Whether the element exists in the list ✓
- c) The length of the list
- d) The position of the head

108. For inserting at any position in a singly linked list, an invalid position message is shown when:

- a) Position == 1
- b) Position == current size
- c) The traversal reaches a NULL node ✓
- d) Position is negative only

109. In insertAtEnd(), if the list is empty, the new node is returned as:

- a) The tail
- b) NULL
- c) The head ✓
- d) The second node

110. When deleting a node at any position, what is done when the position is invalid?

- a) A new node is inserted
- b) The head is deleted
- c) An "Invalid position!" message is printed ✓
- d) The list is cleared

§ SECTION G — Doubly Linked List Operations

111. A Doubly Linked List node in C contains which fields?

- a) data and next
- b) data, prev, and next ✓
- c) data, prev, next, and index
- d) data only

112. When inserting at the beginning of a doubly linked list, the previous head's prev is set to:

- a) NULL
- b) The new node ✓
- c) The tail
- d) head->next

113. Displaying a Doubly Linked List backward requires first traversing to:

- a) The head
- b) The second node

- c) The last node ✓
- d) The middle node

114. When inserting a new node at any position in a doubly linked list and the position is invalid, which action is taken?

- a) Node is inserted at head
- b) Node is inserted at tail
- c) Node is freed and head is returned ✓
- d) Program exits

115. When deleting the front node of a doubly linked list, which pointer must be updated?

- a) tail->prev
- b) new head's prev pointer (set to NULL) ✓
- c) new head's next pointer
- d) No pointer update needed

116. In doubly linked list deletion of the last node, which operation disconnects it?

- a) tail->prev->next = NULL ✓
- b) tail->next = head
- c) head->prev = NULL
- d) tail->prev = NULL

117. In a doubly linked list, backward display starts from:

- a) The first node
- b) The last node ✓
- c) The second node
- d) The middle node

118. In doubly linked list, deleting a middle node requires updating:

- a) Only next pointers
- b) Only prev pointers
- c) Both next and prev pointers of neighboring nodes ✓
- d) No pointer updates

119. A Doubly Linked List is preferable over a Singly Linked List when:

- a) Memory is limited
- b) Forward-only traversal is needed
- c) Backward traversal is needed ✓
- d) Elements are always sorted

120. In doubly linked list node creation, newNode->prev is initially set to:

- a) head
- b) tail
- c) NULL ✓
- d) newNode itself

§ SECTION H — Circular Linked Lists

121. The key property that distinguishes a Circular Linked List from a regular one is:

- a) Nodes have two pointers
- b) The last node points to the head instead of NULL ✓
- c) Nodes are stored in contiguous memory
- d) It has no head node

122. Which of the following is an advantage of a Circular Linked List?

- a) Easier to debug
- b) No null pointers / references ✓
- c) Less memory usage
- d) Supports random access

123. Which of the following is a disadvantage of a Circular Linked List?

- a) No dynamic size
- b) Cannot traverse forward
- c) Infinite loop risk ✓
- d) Cannot delete nodes

124. Which application commonly uses Circular Linked Lists?

- a) Binary Search
- b) Round Robin Scheduling ✓
- c) AVL Trees
- d) Hash Tables

125. In a Circular Singly Linked List, traversal terminates when:

- a) `temp == NULL`
- b) `temp->next == NULL`
- c) `temp == head` (back to start) ✓
- d) `temp->data == 0`

126. When inserting at the beginning of a Circular Singly Linked List, after creating the new node, the last node's next must point to:

- a) NULL
- b) The old head
- c) The new node (which becomes new head) ✓
- d) The second node

127. In creating a Circular Singly Linked List, the first node's next is set to:

- a) NULL
- b) The second node
- c) The head itself ✓

- d) The last node

128. When deleting the beginning node of a Circular Singly Linked List with multiple nodes, which step maintains circularity?

- a) head = NULL
- b) last->next = head (new head) ✓
- c) head = head->prev
- d) last->next = NULL

129. When deleting the last node of a Circular Singly Linked List, the second-to-last node's next must point to:

- a) NULL
- b) Itself
- c) The head ✓
- d) The deleted node

130. Which function is used to count nodes in a Circular Singly Linked List?

- a) size()
- b) length()
- c) countNodes() ✓
- d) getSize()

131. In a Circular Doubly Linked List, the head's prev pointer points to:

- a) NULL
- b) The second node
- c) The last node ✓
- d) Itself

132. In a Circular Doubly Linked List, the last node's next pointer points to:

- a) NULL
- b) Itself
- c) The head ✓
- d) The second-to-last node

133. When inserting a node at the end of a Circular Doubly Linked List, last = head->prev gives:

- a) The second node
- b) The middle node
- c) The current last node ✓
- d) The head

134. Which traversal pattern is used for displaying a Circular Doubly Linked List backward?

- a) Starting from head, using next
- b) Starting from head->prev, using prev ✓
- c) Starting from head, using prev

- d) Starting from head->next, using prev

135. For a Circular Doubly Linked List with only one node, both next and prev point to:

- a) NULL
 - b) The head node itself ✓
 - c) The last inserted node
 - d) NULL and head respectively
-

§ SECTION I — Polynomial Representation using Linked Lists

136. A polynomial term is made up of:

- a) Variables only
- b) Coefficient and exponent ✓
- c) Coefficients only
- d) Exponents only

137. In the polynomial $P(x) = 4x^3 + 6x^2 + 7x + 9$, the degree is:

- a) 4
- b) 2
- c) 9
- d) 3 ✓

138. Array representation of polynomials assumes exponents are arranged from:

- a) Highest to lowest
- b) Random order
- c) 0 to the highest value ✓
- d) Alphabetical order

139. Which representation is preferred for sparse polynomials?

- a) Array
- b) Linked List ✓
- c) Stack
- d) Queue

140. In polynomial representation using a linked list, each node contains:

- a) Coefficient only
- b) Exponent only
- c) Coefficient, exponent, and next pointer ✓
- d) Coefficient, exponent, prev, and next pointers

141. Which is a disadvantage of array representation of polynomials?

- a) Simple storage
- b) Easy retrieval
- c) Waste of space for sparse polynomials ✓

- d) Good for all polynomial types

142. Which is a disadvantage of linked list representation of polynomials?

- a) Uses more memory for dense polynomials
- b) Hard to add two polynomials
- c) Cannot traverse backward through the list ✓
- d) Requires fixed size

143. In polynomial addition using linked lists, when exponents of two terms are equal:

- a) Copy either term to result
- b) Skip both terms
- c) Create a new node with the sum of coefficients ✓
- d) Multiply the coefficients

144. In polynomial addition, if exponent of $p_1 >$ exponent of p_2 :

- a) Copy node of p_2 to result
- b) Skip both nodes
- c) Copy node of p_1 to result ✓
- d) Add both coefficients

145. In polynomial multiplication using linked lists, the exponents of multiplied terms are:

- a) Subtracted
- b) Kept the same
- c) Added ✓
- d) Multiplied

§ SECTION J — Comparison & Conceptual Questions

146. Which sorting/searching algorithm CANNOT be applied to a Linked List efficiently?

- a) Linear Search
- b) Binary Search ✓
- c) Bubble Sort
- d) Insertion Sort

147. Which data structure uses a "roll of paper or tape" as an analogy for access?

- a) Array
- b) Stack
- c) Linked List ✓
- d) Heap

148. Which data structure uses a "book" as an analogy for access?

- a) Linked List
- b) Stack
- c) Array ✓

- d) Queue

149. For dynamic size requirements, which is more suitable?

- a) Arrays
- b) Linked Lists ✓
- c) Static arrays
- d) BSS memory

150. Which of the following linked list types supports bidirectional traversal?

- a) Singly Linked List
- b) Circular Singly Linked List
- c) Doubly Linked List ✓
- d) None of the above

151. Applications of lists include which of the following?

- a) Basis for stacks and queues ✓
- b) Implementation of hardware drivers
- c) Direct memory access operations
- d) Process scheduling tables

152. In which real-world application is a Circular Linked List most naturally used?

- a) Binary Search Tree traversal
- b) Multi-player game turns ✓
- c) Sorting algorithms
- d) Hash table chaining

153. Which type of linked list is used in Image/Slide Show Viewers?

- a) Singly Linked List
- b) Stack
- c) Circular Linked List ✓
- d) Doubly Linked List only

154. Round Robin Scheduling in operating systems commonly uses which data structure?

- a) Array
- b) Binary Tree
- c) Circular Linked List ✓
- d) Hash Table

155. Which memory allocation type is less efficient according to the course material?

- a) Dynamic memory allocation
 - b) Heap allocation
 - c) Static memory allocation ✓
 - d) Runtime memory allocation
-

§ SECTION K — C Programming & Implementation MCQs

156. In C, which header is required for using malloc(), calloc(), realloc(), and free()?

- a) <stdio.h>
- b) <string.h>
- c) <stdlib.h> ✓
- d) <math.h>

157. In a singly linked list program, when malloc() returns NULL, it indicates:

- a) The node is empty
- b) Memory allocation failed ✓
- c) The list is full
- d) The pointer is invalid

158. What does exit(1) do in a C program?

- a) Pauses the program
- b) Terminates the program with an error status ✓
- c) Returns to main
- d) Frees all memory

159. #define MAX 5 in C defines MAX as:

- a) A variable of value 5
- b) A macro constant with value 5 ✓
- c) An integer array of size 5
- d) A pointer to 5

160. In the singly linked list insertAtEnd function, if head == NULL, the function returns:

- a) NULL
- b) A new node as the new head ✓
- c) The old head
- d) -1

161. In the singly linked list deleteFront function, head = head->next achieves:

- a) Inserting at front
- b) Moving head to the next node ✓
- c) Deleting the last node
- d) Clearing the list

162. In a doubly linked list deleteFront function, after moving head forward, the new head's prev must be set to:

- a) The old head
- b) NULL ✓
- c) The tail
- d) Itself

163. In the search function for a singly linked list, the variable position starts at:

- a) 0
- b) 1 ☒
- c) -1
- d) The list size

164. In the circular singly linked list display() function, a do-while loop is used because:

- a) The loop must run at least once (starting from head) ☒
- b) The while loop doesn't work for circular lists
- c) do-while is faster
- d) The number of nodes is always 1

165. In the singly linked list deleteAtPosition(), before freeing a node, its predecessor's next is set to:

- a) NULL
- b) The predecessor itself
- c) The node to be deleted
- d) The successor of the deleted node ☒

§ SECTION L — Fill-in-the-Blank Style MCQs

166. The formula for the position to insert at the end of a list is: Position = _____

- a) Current Size
- b) Current Size - 1
- c) Current Size + 1 ☒
- d) Max Size

167. In BSS segment, all static and global variables are initialized to _____ by the kernel.

- a) 1
- b) -1
- c) Garbage value
- d) Arithmetic 0 ☒

168. A data structure is a _____ of organizing data.

- a) Language
- b) Hardware
- c) Particular way ☒
- d) Random method

169. The calloc() function is different from malloc() because it initializes memory to _____ and takes _____ parameters.

- a) Garbage value, one
- b) 0, two ☒

- c) 1, three
- d) NULL, one

170. In C, sizeof(int) typically returns _____ bytes.

- a) 1
- b) 2
- c) 4 ☒
- d) 8

171. The head of a linked list is a pointer to the _____ node.

- a) Last
- b) Middle
- c) First ☒
- d) Random

172. In polynomial addition, when exponent of $p_1 < \text{exponent of } p_2$, we copy node of _____ to the result.

- a) p_1
- b) p_3
- c) p_2 ☒
- d) Neither

173. A _____ is a user-defined data type which encapsulates a range of data values and their functions.

- a) Primitive data type
- b) Abstract Data Type ☒
- c) Pointer
- d) Structure

174. In a Circular Singly Linked List, when the list has only one node, that node's next points to _____.

- a) NULL
- b) The second node
- c) Itself (head) ☒
- d) The tail

175. Static memory allocation uses the _____ data structure for management.

- a) Queue
 - b) Heap
 - c) Stack ☒
 - d) Tree
-

§ SECTION M — True/False-Style MCQs (Choose the Correct Statement)

176. Which statement about ADT is TRUE?

- a) ADT depends on a specific implementation
- b) ADT is independent of any particular implementation ✓
- c) ADT cannot describe operations
- d) ADT is only for primitive types

177. Which statement about arrays is TRUE?

- a) Arrays support dynamic resizing
- b) Array elements are stored at non-contiguous locations
- c) Array elements are stored at contiguous memory locations ✓
- d) Arrays cannot be searched linearly

178. Which statement about linked lists is TRUE?

- a) Linked lists support random access
- b) Linked lists require no extra memory for pointers
- c) Linked lists allow dynamic sizing ✓
- d) Linked lists are always faster than arrays

179. Which statement about malloc() is TRUE?

- a) malloc() initializes memory to 0
- b) malloc() takes two parameters
- c) malloc() returns a void pointer ✓
- d) malloc() is used for static allocation

180. Which statement about free() is TRUE?

- a) free() reallocates memory
- b) free() initializes memory to 0
- c) free() automatically runs after every malloc()
- d) free() de-allocates dynamically allocated memory ✓

181. Which statement about Circular Linked Lists is TRUE?

- a) The last node always points to NULL
- b) Circular linked lists risk infinite loops if not handled correctly ✓
- c) Circular linked lists use less memory than singly linked lists
- d) Circular linked lists do not have a head node

182. Which statement about the Doubly Linked List is TRUE?

- a) It has only one pointer per node
- b) Traversal is only forward
- c) It has both prev and next pointers in each node ✓
- d) It cannot be used for deletion

183. Which statement about heap memory is TRUE?

- a) Heap size is fixed
- b) Heap is only accessible via arrays
- c) Heap grows in the same direction as stack
- d) Heap memory can change during runtime ✓

184. Which statement about the Text Segment is TRUE?

- a) It has read-write permission
- b) It stores global variables
- c) It is read-only and shareable ✓
- d) It stores dynamically allocated data

185. Which statement about Non-Primitive Data Structures is TRUE?

- a) They cannot be user-defined
- b) They are always Linear
- c) They can be classified as Linear and Non-Linear ✓
- d) They are identical to primitive data structures

§ SECTION N — Advanced & Applied MCQs

186. Which of the following best describes the Time Complexity of accessing an element in an array by index?

- a) $O(n)$
- b) $O(\log n)$
- c) $O(n^2)$
- d) $O(1)$ ✓

187. Which of the following best describes the Time Complexity of accessing an element in a singly linked list by position?

- a) $O(1)$
- b) $O(n)$ ✓
- c) $O(\log n)$
- d) $O(n^2)$

188. When inserting at the beginning of an array-based list, what is the time complexity due to shifting?

- a) $O(1)$
- b) $O(\log n)$
- c) $O(n)$ ✓
- d) $O(n^2)$

189. Which operation on a singly linked list is $O(1)$ when only the head pointer is maintained?

- a) Insertion at the end

- b) Deletion from the end
- c) Insertion at the beginning ✓
- d) Searching

190. The Circular Doubly Linked List deletion at the end uses which pointer access to find the last node?

- a) head->next
- b) head->prev ✓
- c) head->next->next
- d) head->data

191. In a Circular Doubly Linked List insert at end, last = head->prev means:

- a) Accessing the first node
- b) Accessing the second node
- c) Accessing the last node ✓
- d) Accessing the middle node

192. When realloc() shrinks memory, the data in the removed portion is:

- a) Moved to a safe location
- b) Initialized to zero
- c) Lost ✓
- d) Stored in BSS

193. A linked list implementation of a polynomial is preferred because:

- a) It requires more space for dense polynomials
- b) It avoids the need for large arrays for sparse polynomials ✓
- c) It allows random access to terms
- d) It uses less code

194. Which segment contains values of all global variables that are NOT initialized?

- a) Data Segment
- b) Text Segment
- c) BSS (Uninitialized Data Segment) ✓
- d) Stack

195. Which function can be used to change the size of previously allocated memory?

- a) malloc()
- b) calloc()
- c) realloc() ✓
- d) free()

196. In a singly linked list, deleting the last node requires traversal of:

- a) Only the head
- b) The entire list until the second-to-last node ✓
- c) Only the tail

- d) No traversal

197. The do-while loop is used in circular linked list traversal to ensure:

- a) The list can be traversed backward
- b) The head node is always processed at least once ✓
- c) Infinite traversal is avoided automatically
- d) The loop runs only once

198. In a Circular Doubly Linked List with only one node, deleting that node results in:

- a) An empty list (head = NULL) ✓
- b) A list with one remaining node
- c) A segmentation fault
- d) The node pointing to itself

199. Which of the following is true about inserting into a doubly linked list at the end?

- a) Only the new node's next needs to be updated
- b) Only head->prev needs to be updated
- c) Both the last node's next and the new node's prev must be updated ✓
- d) No pointer updates are needed

200. What happens when an invalid position (beyond list size) is given to deleteAtPosition() in a singly linked list?

- a) The last node is deleted
- b) The program crashes
- c) An "Invalid position!" message is printed and the list is unchanged ✓
- d) The head is deleted

§ SECTION O — Bonus Questions (201–215)

201. Which of the following operations is NOT directly supported by the List ADT?

- a) get()
- b) insert()
- c) sort() ✓
- d) isEmpty()

202. The removeAt() operation in a List ADT removes the element at:

- a) The front
- b) The end
- c) A specified location ✓
- d) A random location

203. Which type of memory allocation is described as "Variables get allocated permanently"?

- a) Dynamic

- b) Heap
- c) Static ✓
- d) Runtime

204. The isFull() operation returns TRUE when:

- a) Current Size == 0
- b) Current Size == Max Size ✓
- c) Current Size > Max Size
- d) Current Size == 1

205. When inserting at position n in a list, elements from position n to Current Size are shifted:

- a) One position backward
- b) One position forward ✓
- c) Two positions forward
- d) Not shifted

206. A node deletion in any linked list always requires calling _____ to prevent memory leaks.

- a) malloc()
- b) calloc()
- c) realloc()
- d) free() ✓

207. In a linked list traversal, when does the loop terminate for a singly non-circular list?

- a) When temp == head
- b) When temp->data == 0
- c) When temp == NULL ✓
- d) When temp->next == head

208. What is the structure definition keyword used to define linked list nodes in C?

- a) class
- b) union
- c) struct ✓
- d) typedef only

209. The initialized data segment stores:

- a) Only local variables
- b) External, global, static, and constant variables whose values are initialized ✓
- c) Code instructions
- d) Dynamically allocated data

210. Which linked list type is most suitable for implementing a circular queue?

- a) Singly Linked List
- b) Doubly Linked List

- c) Circular Linked List ✓
- d) Non-circular Doubly Linked List

211. In a doubly linked list node inserted at any position, if the next node exists, its _____ must be updated.

- a) next pointer
- b) data field
- c) prev pointer ✓
- d) head pointer

212. Token Passing in Networks (Ring Topology) uses:

- a) Stack
- b) Queue
- c) Circular Linked List ✓
- d) Binary Tree

213. Polynomial addition using arrays is described as:

- a) Complex and inefficient
- b) Straightforward, adding element-wise from 0 to n-1 ✓
- c) Not possible
- d) Identical to linked list method

214. Which of the following memory allocation functions DOES NOT initialize the allocated memory to any specific value?

- a) calloc()
- b) realloc() (for new blocks only)
- c) malloc() ✓
- d) Both a and c

215. A linked list where traversal from the last node can return to the first node without resetting the pointer is called:

- a) Doubly Linked List
- b) Singly Linked List
- c) Circular Linked List ✓
- d) Static Linked List